Apply Area Concepts to Solve Problems

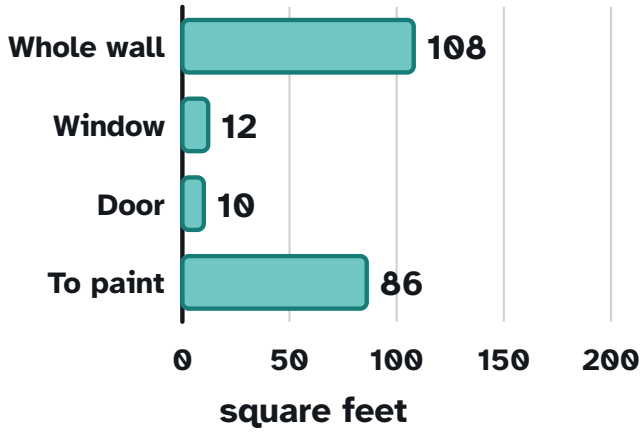
Decide for each piece of a shape whether to add it or subtract it.

Name _____ **Date** _____

Tonight's spark

A painter will paint a wall that is 12 feet wide and 9 feet tall. The wall has a window that is 3 feet by 4 feet and a door that is 2 feet by 5 feet. How many square feet need paint?

Guess first: the whole wall is 108 square feet. Will the painted part be more or less?



Our class model: part of it or missing from it

Try it

Work it out on paper. Write one answer on each line.

1 **A piece that has been CUT OUT of a figure gets what?**

A. Subtracted **B.** Added **C.** Multiplied **D.** Doubled

My answer _____

2 **A 7 by 5 rectangle has a 3 by 2 rectangle joined to its side. What is the total area?**

A. 29 square units **B.** 35 square units **C.** 41 square units **D.** 12 square units

My answer _____

3 **A 10 by 6 rectangle has a 4 by 3 hole. What is the area of what is left?**

A. 72 square units **B.** 48 square units **C.** 12 square units **D.** 60 square units

My answer _____

4 **A house shape is an 8 by 5 rectangle with a triangle roof on top. The roof has a base of 8 and a height of 3. What is the total area?**

A. 64 square units **B.** 28 square units **C.** 43 square units **D.** 52 square units

My answer _____

Talk about it

Which piece did you add, which did you subtract, and how did you decide?

Home mission

Look at a wall at home. Pretend it is 10 feet wide and 8 feet tall with one window that is 3 feet by 3 feet. How much wall needs paint? Which piece do you subtract?

If we get stuck, ask

- Is this piece part of the figure, or missing from it?
- What simple shapes can you break this into?
- Did every piece get counted exactly once?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/5-4-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it